

13. Answer: B

The total gravitational field strength g due to both X and Y can be determined by getting the vector sum of the individual gravitational field strengths g_x and g_y due to X and Y respectively.

From the vector sum, it can be seen that the value of g will decrease from point P, and reach $g = 0$ at the midpoint of PQ, which is also the midpoint of XY. This is because at this midpoint, g_x and g_y are opposite in direction and equal in magnitude. The value of g will then increase it approaches point Q.

14. Answer: C